3. Apply the IIoT. It provides data, which when analysed gives detailed information about the machine's running state.

4. Apply AR as GUI to digital data in a user-understandable format.

Preparing an enterprise for digital connectivity

All industrial revolutions have been driven by innovation and technology. For the first industrial revolution, mechanisation was the driver and steam, the enabler.

For the second industrial revolution, mass production was the driver and electricity, the enabler.

For the third revolution, PCs, connectivity and automation were the drivers and IT, the enabler.

For the fourth revolution, the IoT, artificial intelligence (AI), cloud, robotics, AR, VR, mixed reality (MR) and chatbots are the drivers, and connectivity, the enabler. The fourth industrial revolution is powered by connectivity, and has enabled expanding the scope of business-critical applications. To unlock this, it is important to become adept at controlling the physical with digital means.

Success of industrial automation depends on reliability, connectivity, distributed intelligence, transparent IT architecture, security, easy integration, flexible transport, additive manufacturing, standards and software services.

The digital value chain for Industry 4.0 can be split into seven key components, as follows:

1. Machines/devices. Ecosystem-based approach, leveraging on existing deployment

2. Network. Multi-access network management, proposing the best technology

3. Platforms. IT infrastructure-interface enablers, platforms to nurture ubiquitous approach

4. Applications. Industry-specific use-case, machine learning and analytics

5. Solutions. Business ease creation, solution design and delivery

6. Operations. Project implementation and support

7. Project management. Maintaining network and applications uptime, periodic upgrades for security and operational improvement

Connectivity is paving the way for such industries as automotive, oil and gas, banking and finance, IT, power and production, to be ready for Industry 4.0. There are verticals like real-estate, healthcare, project development, trading, aviation and others adopting LTE in this direction.

Some industrial automation use-cases

Nokia and BSNL have collaborated for Industry 4.0, and have moved from traditional wired factory networks to wireless networks, leading to improved equipment mobility for enhanced flexibility of manufacturing infrastructure. Early adoption of IIoT applications in industries have improved productivity and operational efficiency, too.

Rockwell Automation's transformation results have been as follows: delivery gone up from 82 to 90 per cent, lead times reduced by fifty per cent, productivity increased more than five per cent and thirty per cent reduction in capex.

PTC AR by Vuforia studio provides powerful AR content creation for enterprise content creators and publishing solutions for industrial enterprises. Vuforia engine has use-cases in inspection and manufacturing operations, marketing and sales, pre-sales manufacturing process services, training and project reviews.

Surface inspection systems for digital TV production by Samsung reduce human operator intervention in the production line, increase throughput and reduce errors in product inspection. System operation supports capturing TV surface image from high-resolution camera devices.

Samsung has also developed a post-processing system for an image captured by the camera to make it amenable to automatic detection. Automatic surface defect detection algorithm on a pre-processed image enables distinguishing of a real defect from a false one on the digital TV.

The connected cyber physical world has brought availability of information and ease of dissemination of information. Information-driven enterprises have enabled optimisation of resources. Leakage or waste can be minimised through connectivity, technology and transparency. For example, a 45kg coal wagon going from Jharkhand to NTPC plant results in at least thirty per cent wastage in resources. With the right technology connecting the point of source to the point of use, NTPC has to pay only for the coal that lands at the plant. This can be achieved through weighing the wagon at different locations through the whole passage, and this information connectivity can reduce under-invoicing and over-invoicing. This ultimately benefits the end consumer, reducing the price of per unit electricity.

Government and industry initiatives

By anticipating challenges before they arise is more likely to offer a successful digital transformation. Challenges experienced are lack of understanding, lack of scalability and poor workforce skills. Driving the next industrial revolution by reinventing 3D can help achieve immersive and easier designs and prototypes, and the final part of digital production.

Government initiatives for fuelling manufacturing leadership and innovations redefine and create new value opportunities.

To strengthen Make in India ecosystem with the adoption of smart technologies, Automation Industry Association and research institutes like IIT-Delhi and others have created a special-purpose vehicle, called Foundation for Smart Manufacturing (FSM), to take the emerging wave of smart technologies and adapt it with relevance to the needs of the Indian industry.

To facilitate, introduce and enlighten the industry with an evolved genre of quality-focused smart manufacturing, Department of Heavy Industry (DHI) supported by IITD-AIA Foundation for Smart Manufacturing (FSM), Federation of Indian Chambers of Commerce & Industry (FICCI) and IIoT India has been organising workshops to empower different industrialists to achieve the benefits of Industry 4.0.

Digitisation in the industrial manufacturing segment is about using capacity and a properly-calibrated information-driven solution to solve problems and reduce pain-points in the enterprise. And, early adopters of technology always benefit more by the industrial revolution digitisation. EFY